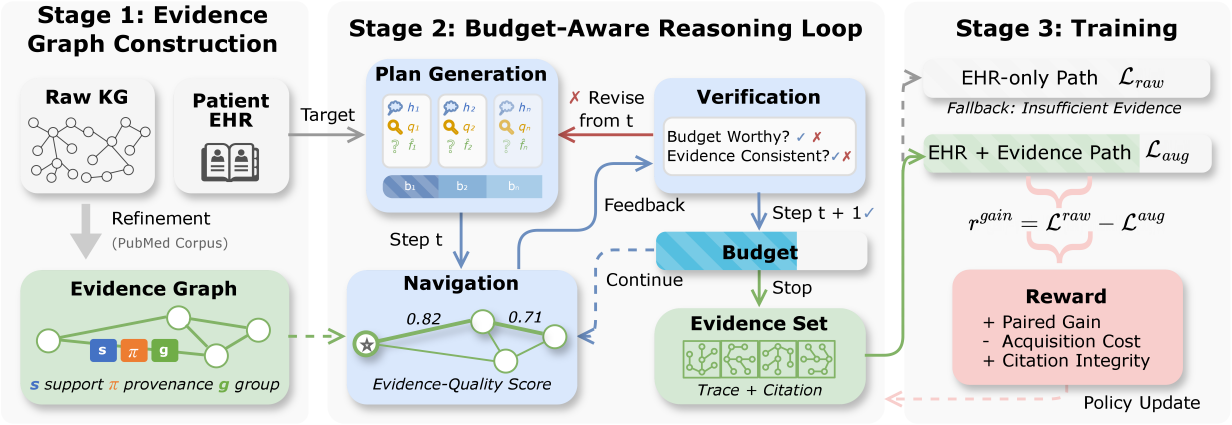

Stage 1: Evidence Graph Construction
Raw KG
Patient EHR
Refinement
(PubMed Corpus)
Evidence Graph
s support π provenance g group
Stage 2: Budget-Aware Reasoning Loop
Target
Plan Generation
Revise from t
Verification
Budget Worthy? ✓ ✗
Evidence Consistent? ✓ ✗
Feedback
Step t + 1 ✓
Step t
Budget
Continue
Stop
Navigation
0.82
0.71
Evidence-Quality Score
Evidence Set
Trace + Citation
Stage 3: Training
EHR-only Path $\mathcal{L}_{raw}$
Fallback: Insufficient Evidence
EHR + Evidence Path $\mathcal{L}_{aug}$
$r^{gain} = \mathcal{L}^{raw} - \mathcal{L}^{aug}$
Reward
+ Paired Gain
- Acquisition Cost
+ Citation Integrity
Policy Update